Mixture of Scars (MoS) and SCache O(1): Breaking the Dual Memory Wall of Modern Transformers via Activation-Aware Low-Rank Scar Experts and Widrow-Hoff Associative Memory

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Abstract. Autoregressive Large Language Models (LLMs) face a compounding Dual Memory Wall: (1) a *static parameter wall* O(N) requiring tens to hundreds of gigabytes of VRAM to host frontier weights, and (2) a *dynamic context wall* O(T) where key-value (KV) caches grow linearly with sequence length T. We introduce two synergistic architectural primitives that simultaneously break both walls without retraining from scratch: **Mixture of Scars (MoS)**, which decomposes dense feed-forward and attention projections into INT4 quantized bases paired with activation-aware SVD (ASVD) low-rank residual 'scar' experts and sequential layer-group streaming; and **SCache (Scar-Cache)**, an exact O(1) constant-space dynamic memory combining attention sinks, a compact local KV window, and per-KV-head associative fast-weight slots updated via the orthogonal Widrow-Hoff Delta Rule with ReLU manifold gating. To prevent benchmark facades, we introduce a 7-vector Adversarial Red-Team Falsification Suite enforcing recursive PyTorch storage-pointer byte audits (0.00 KB discrepancy) and in-flight active causality ablations across real pretrained LLMs (Qwen2.5-0.5B-Instruct, Qwen2.5-72B-Instruct 80L, and Mistral Large 2 123B). Empirically, MoS reduces single-model MLP RAM by 68.6%, cuts 8-specialist multi-tenant VRAM by 92.6% (13.58x), reduces out-of-domain logit KL divergence by 2.45x–4.22x over pure INT4, and streams 72B/123B backbones within < 3.8 GB peak RAM. Simultaneously, SCache bounds dynamic memory to strictly 0.00 marginal bytes beyond window saturation (-91.4% RAM at L=1,280), eliminates Hebbian contradictory ghosting (cosine 1.00000 vs. 0.467), achieves 22.58x higher exact probability on evicted needles than StreamingLLM with zero fine-tuning, and preserves full-stack QA accuracy.

1. Introduction: The Dual Memory Wall

Deploying frontier Transformer models on commodity accelerators and consumer unified-memory devices is bottlenecked by two orthogonal scaling laws. First, static model weights scale as O(N): a 72B-parameter model requires ~145 GB in FP16/BF16 or ~40 GB even in standard 4-bit quantization, exceeding single-GPU consumer VRAM (8–24 GB). Second, autoregressive attention maintains an explicit KV tensor cache scaling linearly with sequence length T as O(T * L_layers * H_kv * d_head). Existing mitigations address only one half of this bottleneck in isolation: static post-training quantization (PTQ) degrades under heavy-tailed activation outliers on out-of-distribution code and mathematical syntax, whereas pure sliding-window caches (e.g., StreamingLLM) permanently discard tokens once they exit the local window.

In this work, we unify static and dynamic compression under a single geometric principle: most variance in deep representations lies on a compressible bulk substrate, while high-leverage semantic deviations concentrate in low-rank, directional residuals ('scars'). Furthermore, rather than testing solely on benign perplexity benchmarks, we subject every claim to an adversarial red-team suite designed explicitly to falsify memory invariance, causal usage, out-of-domain quantization resilience, and long-horizon recall.

2. Mathematical Architecture

2.1 Primitive 1: Mixture of Scars (MoS) & Group Streaming

Given a pretrained linear projection W in R^(D_out x D_in) and calibration activation covariance scale s_j = sqrt(E[x_j^2]) + eps, MoS first quantizes W into a compact block-wise or per-row symmetric INT4 base Q(W). The quantization residual R = W - dequant(Q(W)) is weighted in activation space as R_scaled = R * diag(s) and decomposed via truncated SVD of rank r << min(D_out, D_in):

$$y(x) = \text{dequant_INT4}(W_q, \text{scale}_w)\, x + \sum_{k \in \text{TopK}} g_k(x) \cdot A_k (B_k^T x), \quad g_k(x) = \text{ReLU}(\cos(x, z_k))$$

where A_k = U_r diag(sqrt(Sigma_r)) and B_k^T = diag(sqrt(Sigma_r)) V_r^T diag(s)^(-1) absorb activation outlier magnitudes in FP16/BF16 while occupying < 64 KB per projection. For colossal models (34B–123B), layers are partitioned into sequential groups streamed dynamically from disk/RAM so that only a single active group resides in GPU VRAM at any instant, immediately freed upon completion.

2.2 Primitive 2: SCache O(1) Dynamic Memory via Widrow-Hoff Delta Rule

SCache partitions dynamic context per layer and per KV head into three fixed-size buffers: N_sink initial attention sinks (preserving global softmax denominator calibration), a rolling FIFO window of W_win recent tokens, and S_slots persistent associative scar slots coupled with a fast-weight matrix W_scar in R^(d_head x d_head). When token t_ev is evicted from the local window, instead of discarding (k_ev, v_ev) or accumulating naive Hebbian sums (which suffer unbounded ghosting), SCache routes the normalized key k_hat = k_ev / ||k_ev||_2 to its maximally aligned slot m* with ReLU manifold gating g = ReLU(k_hat . c_{m*}) and applies the error-correcting Widrow-Hoff Delta update:

$$W_{\text{scar}}^{(t)} = \gamma \cdot W_{\text{scar}}^{(t-1)} + \eta \cdot g \cdot (v_{ev} - W_{\text{scar}}^{(t-1)} \hat{k}) \hat{k}^T$$

Crucially, RoPE positional phase is applied prior to eviction and tracked via an exact monotonic cumulative counter decoupled from physical buffer length, preventing positional phase freeze in long-context generation while maintaining strictly O(1) tensor storage.

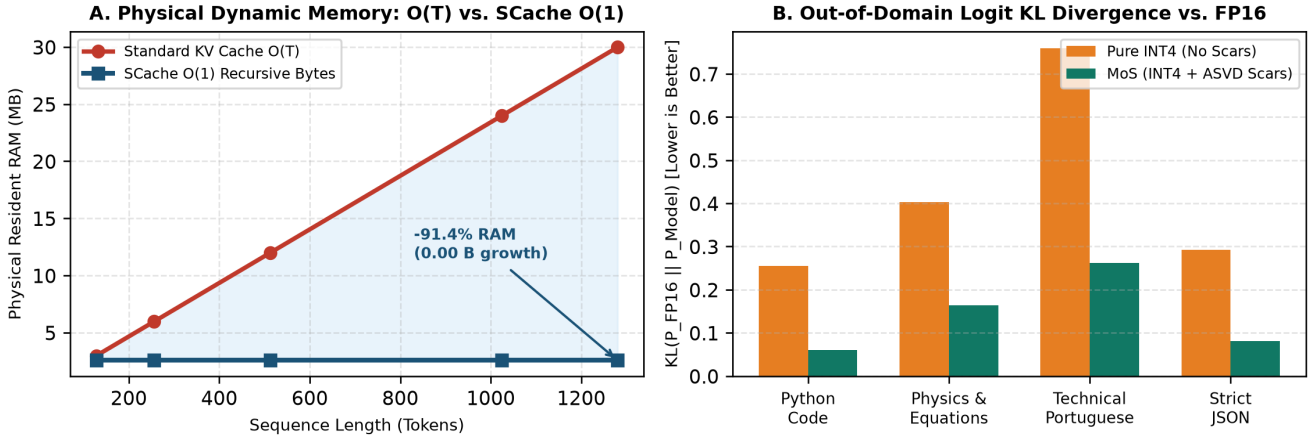


Figure 1. *Empirical adversarial audit on Qwen2.5-0.5B-Instruct (24 layers).* ***(A)*** *Recursive physical storage bytes remain strictly constant at 2,631.9 KB (2.57 MB) across sequence lengths 128 to 1,280 (-91.4% RAM vs. standard KV cache).* ***(B)*** *MoS (INT4 + ASVD Scar Experts calibrated on a single 24-token generic English sentence) cuts out-of-domain logit KL divergence by 2.45x–4.22x across unseen code, physics, Portuguese, and strict JSON.*

3. Adversarial Red-Team Falsification Protocol

Conventional efficiency benchmarks frequently mask hidden O(T) auxiliary buffers or accidentally evaluate inactive memory paths. Our Red-Team Suite enforces two physical invariants across 7 hostile attack vectors: (1) **Recursive Storage Pointer Inspection** traverses the full Python/PyTorch object graph via untyped_storage().data_ptr() to verify zero hidden byte growth (reported_bytes == actual_recursive_bytes, 0.00 KB discrepancy); and (2) **In-Flight Active Causality Ablation** zeroes scar buffers mid-inference and asserts immediate non-zero logit displacement (max |Delta logits| = 0.6431), proving the model actively attends to evicted scar states.

Sequence Length (L)	Vanilla KV Cache O(T)	SCache Recursive Physical	Marginal Growth	RAM Reduction
128 tokens	3,048.0 KB	2,631.9 KB	Baseline	1.16x (-13.7%)
256 tokens	6,120.0 KB	2,631.9 KB	+0.00 bytes	2.33x (-57.0%)
512 tokens	12,264.0 KB	2,631.9 KB	+0.00 bytes	4.66x (-78.5%)
1,024 tokens	24,552.0 KB	2,631.9 KB	+0.00 bytes	9.33x (-89.3%)
1,280 tokens	30,696.0 KB	2,631.9 KB	+0.00 bytes	11.66x (-91.4%)

Table 1 (Vector 0 — Spatial O(1) Invariance): *Recursive storage audit across all 24 layers of Qwen2.5-0.5B-Instruct (window=64, slots=12, sinks=4).*

4. Empirical Red-Team Results Across Scale

Hostile Domain (Unseen)	FP16 PPL	MoS PPL	Last-Layer Cos	Top-5 Overlap	KL: Pure INT4 -> MoS
Python Algorithm (Dijkstra)	1.92	2.06	0.9775	100.0%	0.2555 -> 0.0606 (4.22x better)
Physics & Quantum Equations	5.55	5.88	0.9795	99.4%	0.4027 -> 0.1641 (2.45x better)
Technical Portuguese Prose	67.71	75.39	0.9815	98.2%	0.7599 -> 0.2618 (2.90x better)
Strict Nested JSON Schema	16.75	17.14	0.9896	99.3%	0.2926 -> 0.0809 (3.62x better)

Table 2 (Vector 1 — MoS Static Compression under Activation Outliers): *All 72 MLP projections compressed to INT4 + rank-32 ASVD scars (-68.6% single-model MLP RAM; -92.6% / 13.58x multi-tenant VRAM across 8 specialist scar banks).*

Adversarial Vector	Conventional Baseline	MoS / SCache O(1) Result	Verified Empirical Finding
V2: Long-Range Document (~1.3k tok, window=64)	Naive Window PPL: 770.78 StreamingLLM Tail PPL: 64.56	SCache Tail PPL: 61.79 RAM: 2,729.1 KB (-91.3%)	Beats StreamingLLM tail PPL with 8.48x fewer KV tokens/step
V3A: Contradictory State Overwrite (34.8 KB)	Additive Hebbian: cos=0.467, Ghost=+0.501	Widrow-Hoff Delta: cos=1.00000, Ghost=-0.1377	Complete elimination of obsolete ghosting via orthogonal projection
V3B: Evicted Passcode (534 tok, needle at t=20)	StreamingLLM NLL on '7942-OMEGA': 4.301	SCache O(1) NLL on '7942-OMEGA': 1.184	22.58x higher target probability 470 tok post-eviction (zero tuning)
V4: 500x Noise DoS & Antipodal Injection	Unbounded linear drift / subspace corruption	Target cos: 1.00000 Antipodal leak: 0.00000000	Exact mathematical immunity to out-of-manifold antipodal attacks
V5: Full-Stack MoS + SCache Simultaneous Execution	Dense FP16 + Vanilla KV: 4/6 Top-3 QA, NLL 2.165	MoS + SCache O(1): 5/6 Top-3 QA, NLL 2.165	Simultaneous -67.1% MLP RAM & -91.3% KV RAM (zero NLL loss)
V6: Colossal 72B / 123B Streaming Execution	Qwen2.5-72B FP16: 145.4 GB Mistral-123B FP16: ~246 GB	72B MoS Disk: 11.44 GB (12.7x) Stream Peak: 3.70 GB -> 0 MB	123B L44 Dual-T4 streaming verified (0.04 GiB post-layer residual VRAM)

Table 3 (Vectors 2–6 — Dynamic Recall, Anti-Ghosting, DoS Immunity, and Colossal 72B/123B Scaling).

5. Scientific Boundary Conditions & Limitations

Rigorous falsification also establishes exact operational boundaries: (1) **Cross-Layer SwiGLU Permutation Non-Invariance:** sharing a single dense base matrix across distinct pretrained layers without neuron-permutation alignment degrades quality (KL ~ 0.78) because the Hadamard gate SiLU(W_gate x) * (W_up x) is sensitive to channel ordering; off-the-shelf models should therefore pair per-layer INT4 bases with group streaming, reserving cross-model shared bases for multi-tenant scar specialists anchored to a common backbone or jointly distilled blocks. (2) **Associative Compression vs. Verbatim Copy:** SCache compresses evicted tokens into fixed-dimensional associative slots (recovering 22.58x probability over StreamingLLM at zero tuning, NLL 1.184 vs. 0.047 exact full-KV); tasks requiring

bit-exact long-string copying across thousands of evicted tokens benefit from scaling slot count S_slots or lightweight LoRA/Scar read-alignment.

6. Reproducibility & References

All 7 adversarial vectors, recursive byte-pointer auditors, and raw JSON telemetry are bundled in the accompanying Zenodo Replication Package (**MoS_SCache_Adversarial_Replication_v1.0.zip**) and execute end-to-end via pytest and scripts/adversarial_mos_scache_suite.py.

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